

causes a lot of problems for the eyes like fatigue and might be a major cause behind VR sickness as well.

At SIGGRAPH (Special Interest Group on GRAPHics and Interactive Techniques) 2015, Professor Gordon Wetzstein and his colleagues from Stanford demonstrated a new VR headset that accommodates for the vergence-accommodation conflict. It does so by shunning flat images and going for light fields, which are 3D patterns of light rays that mimic light bouncing off objects in the real world. This technology is achieved by using a stack of two LCDs with focussed details for distant objects being on the rear LCD. The division of the image is computed by an algorithm, and according to Wetzstein, that is where the real difference comes in. This device isn't coming out anytime soon, but it has shown that it can be done, cheap.



Motion blur in real life is caused due to an object moving faster than the rate of its image being refreshed by our brain

Another problem is being tackled in an Indian Startup's approach to VR. The phenomenon of judder and smear are very persistent in normal VR. We first need to understand these phenomena to realise its impact on VR and the required solution. In real life, when our eyes are looking at a moving object, or there is any movement relatively between the eye and the object, the object transitions smoothly. But on a display, the object can only transition one frame at a time. So the movement of an object can only be depicted one frame at a time. Imagine this as a kind of stuttered movement